



UNIVERSITY
OF COLOGNE

WORKSHOP 03

Advanced Analytics and Applications

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Coding Tasks

Hard Clustering

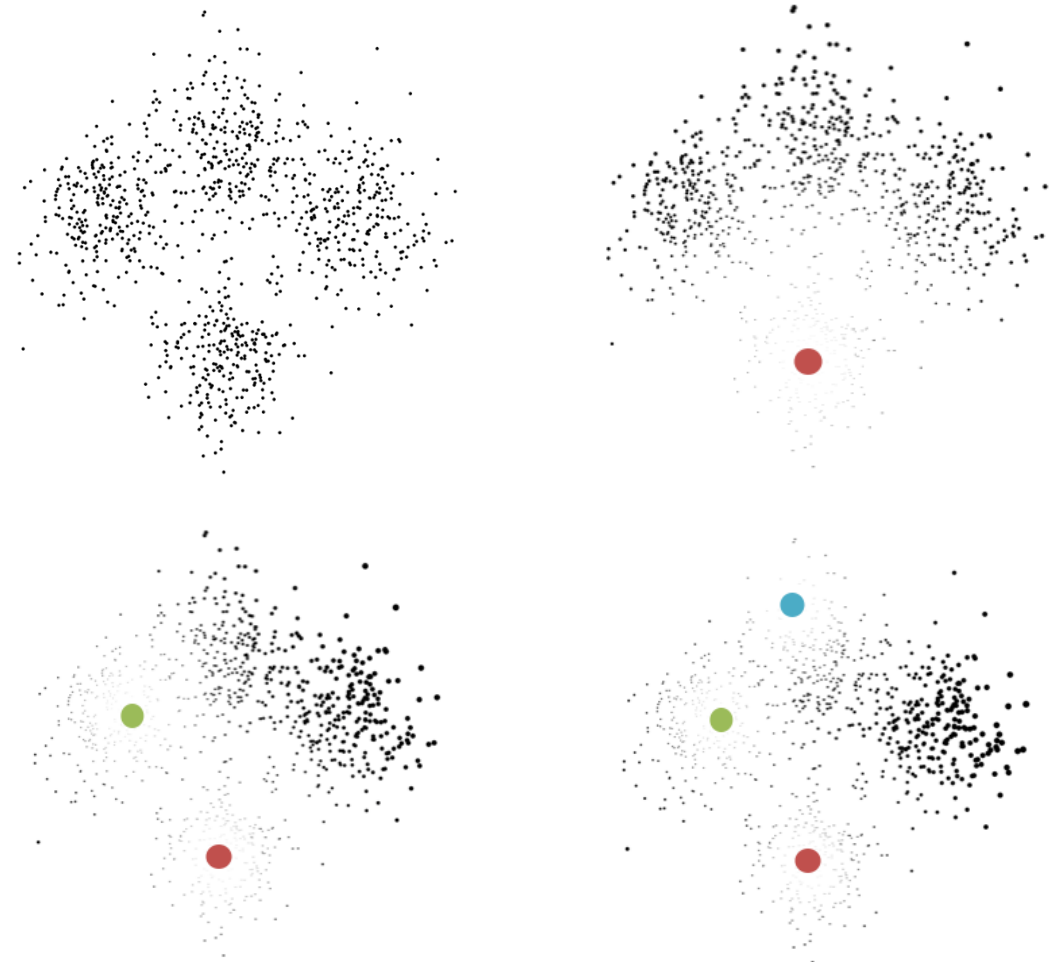
- **Goal:** Partition data into natural groups that are homogeneous within and heterogeneous between each other: Clusters.
 - **Intuition:**
 - Homogeneous: Minimize within-cluster distance between members
 - Heterogeneous: Maximize between cluster (center) distances
 - We can condense the problem to finding centers μ that minimize the sum of all within-cluster distances.
 - **Problem:** This is NP-hard
- K-means is an iterative procedure to find k μ :
 1. Initialize k random cluster centers μ
 2. Repeat until convergence:
 - a) Compute the cluster assignment (closest distance) of every data point x given μ
 - b) Re-compute the cluster centers μ' to be at the centroid/mean of all cluster members

Hard Clustering

- K-means has one **vital drawback**: The initial center placement determines the outcome!
- **K-means++** adds clever initialization:
 1. Initialize one random center μ_1

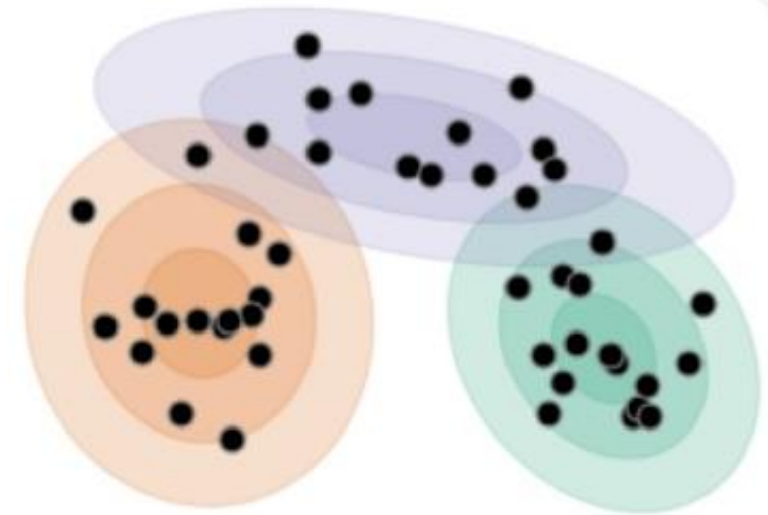
Let $D(x)$ be the distance between all non-chosen points x and the closest center μ .

2. Choose next center μ_2 from $P(x|\mu) \sim D(x)^2$
3. Repeat 2. until k centers are chosen
4. Continue with k-means...



Soft Clustering

- Sometimes we want to have **probabilities** of membership instead of membership assignments
- To find probabilities, we would need to have a **probability function** $p(y|X)$ that gives us the probability of the cluster in question y given X which are all observed features
- The question is, how do we come up with / configure such a probability distribution?



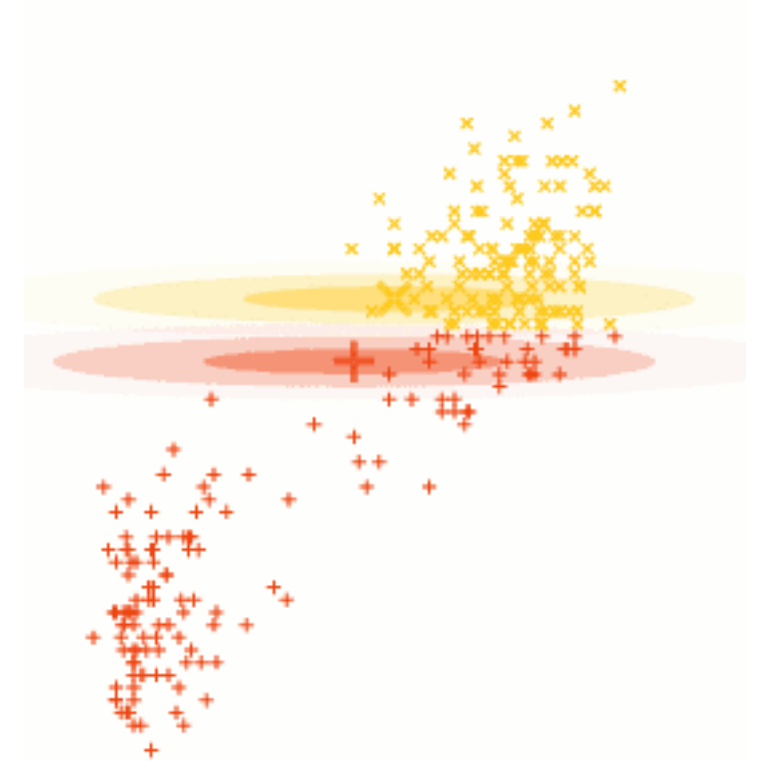
Soft Clustering: GMM

- The short answer is: We construct it ourselves.
- We use **Gaussian Distributions** $\mathcal{N}(\mu, \sigma)$ for probability distributions as they have a nice property of decaying influence towards the limits and are reasonably easy to estimate.
- We use k components $\mathcal{N}(\mu_k, \sigma_k)$ that make up the **mixture model** $p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x | \mu_k, \sigma_k)$, with mixing weights π_k , $\sum \pi_k = 1$. The membership probability is $p(k | x)$ for a component k .
- The longer answer is: Like in k-means we **iteratively test potential candidates** and modify them accordingly. Here, the candidates are not cluster centers however, but distribution coefficients μ, σ and mixing weights π

Soft Clustering: Expectation Maximization

We use as many individual Gaussians (components) k as there are cluster centers and configure them using an iterative algorithm called **Expectation–Maximization**:

1. Start with random initialization of π_k , μ_k and σ_k
2. For each component: Calculate the „probabilities“ $P(y|X, \mu, \sigma)$ that points belong to that cluster, conditional on the current parameters μ, σ – these are latent variables we estimate
3. Tune μ, σ to maximize the likelihood of the model fitting both observed and latent (that have now been estimated) variables



Soft Clustering: Applied

- We finally construct a **soft clustering** by combining all the individual distributions such that every data point can be assigned a **vector of responsibilities**, that represent the weight of influence that a single Gaussian component had in generating this x from the **mixture model** of all the Gaussians combined.

$$\gamma_{nk} = \frac{\pi_k \mathcal{N}(x_n | \mu_k, \sigma_k)}{\sum_i \pi_i \mathcal{N}(x_n | \mu_i, \sigma_i)}$$

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Question 1: Soft Clustering Warm Start

Question: Expectation Maximization starts with random μ, σ, π . Discuss the implications of this and propose a method for warm starting GMM.

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Question: Expectation Maximization starts with random μ, σ, π . Discuss the implications of this and propose a method for warm starting GMM.

Answer:

1. With random initialization, EM can take a lot of time to converge as each iteration is computationally taxing (much more than say k-means). EM is also not guaranteed to converge to a global optimum.
2. We could use hard clustering methods to initialize μ , the "centers" and use the number of data points in each cluster to warm start π , the mixing weights.

Question 2: Outliers in GMM

Question: Your colleague always uses a simple distance threshold in hard clustering (k-means) to exclude data points as outliers that are simply “too far away” from the rest of the data.

- a) Explain to them why their tactic might not work or at least lure them into false conclusions.
- b) Explain to them in simple terms why GMM are more natural when talking about outliers and how you could utilize them for that.

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Answer:

- a) k-means will consider the sum of all distances in determining where to move cluster centers. This means that excluding “far away” outliers afterwards doesn’t consider that they were used like all other data points to determine what “far away” from centers even is.
- b) One intuitive explanation of the independent ($p(x)$ not $p(x|k)$) probability of a GMM is the following: “The probability that an observation x was generated *at all* from the distribution” → Observations below a certain threshold can safely be excluded as outliers as it is not just unlikely that they were generated from one component $p(x|k)$, but from the full mixture!

Multiple Choice Questions: GMM

Question: With an n -dimensional data set (number of features) for finding k clusters...

- ☐ GMM works out of the box, by using multivariate Gaussians
- ☐ You need to have $k(n - 1)$ Gaussian components
- ☐ GMM doesn't work, it only work's with 2-dimensional data

Multiple Choice Questions: GMM

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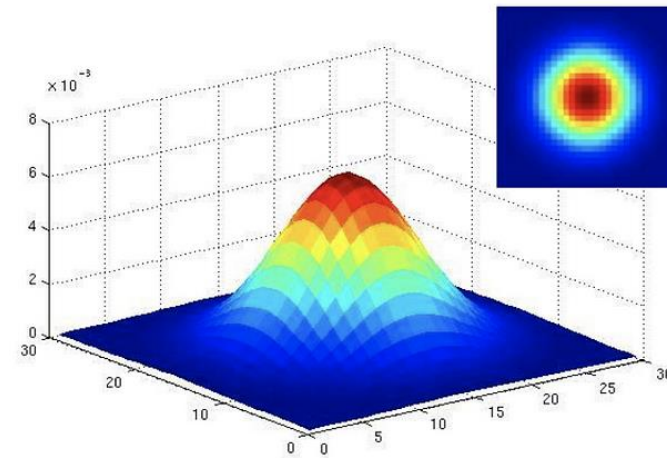
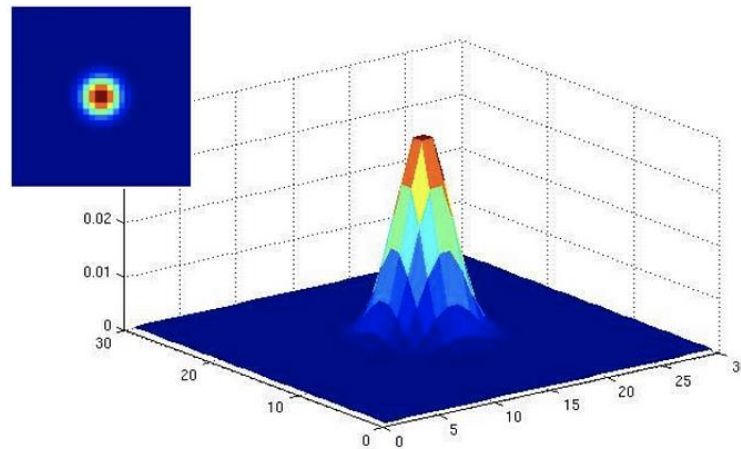
Multivariate Gaussian Distribution:
 $\mathcal{N}(\mu, \sigma)$ with $\mu \in \mathbb{R}^n$ (a vector of size n)
and a covariance matrix Σ of size $n \times n$

Multiple Choice Questions: GMM

Question: The cluster boundaries (considering a threshold) or the „influence boundaries“ of GMM components are always perfectly spherical in 2D.

☐ True

☐ False



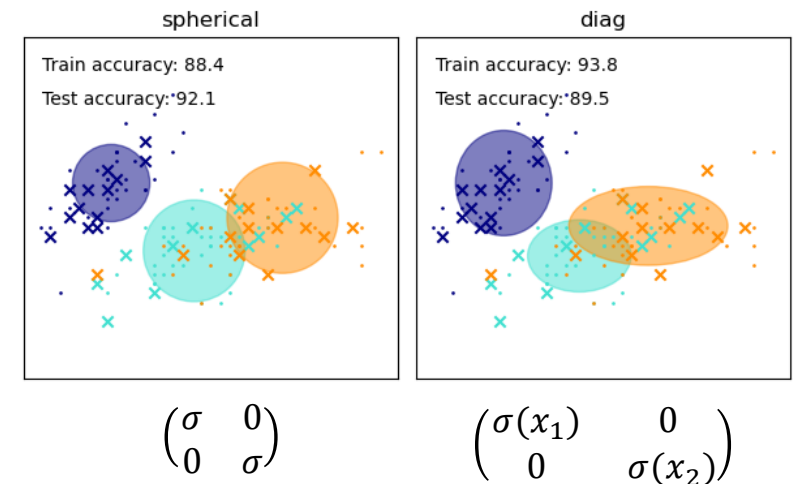
Multiple Choice Questions: GMM

Question: The cluster boundaries (considering a threshold) or the „influence boundaries“ of GMM components are always perfectly spherical in 2D.

☐ True

☒ False

This is only true if the covariance matrix has the same elements on the diagonal. If elements on the diagonal are different, then the variance is different in different dimensions:



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Contact



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